

Kartik Agnihotri

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Summary

Software Engineer with 2+ years of experience building production backend and AI systems using **Node.js, JavaScript, PostgreSQL, Redis and AWS**. Experienced in shipping AI agents, REST APIs, event-driven workflows, OAuth integrations, asynchronous processing, production debugging, performance optimization, security and third-party API integrations. Experienced with **Claude Code, Codex, Gemini and Cursor** for AI-assisted development.

Experience

Full Stack Software Engineer, Fragmatic Technologies

April 2024 – Present

- Designed and owned **5 production REST APIs** for a multi-tenant B2B platform, covering authentication, validation, business logic, **Sequelize/PostgreSQL**, deployment and production support; contributed to **40+ APIs**. Optimized PostgreSQL listing queries, reducing typical response time from **30–40s to 3–8s**.
- Built and operated **event-driven workflows** using **Redis, Bull, AWS SQS, Lambda and API Gateway**, handling asynchronous processing, retries, caching, TTL-based processing and failure recovery. Debugged high-error-rate Lambda/SQS pipelines using CloudWatch and application logs.
- Built a **production agentic AI system in Node.js** end-to-end, from requirements and architecture through implementation, deployment, monitoring, optimization and iteration. Implemented tool/function calling, planning, multi-step execution, subagents, conversational state, structured outputs, schema validation, streaming and human approval for mutations.
- Designed **role-based tool authorization and audit logging** for the AI agent, restricting tools by authenticated user permissions and recording orchestration, subagent, tool, argument and response activity while protecting sensitive data. Implemented JWT/OAuth authentication, API authorization, IAM, secrets management and encryption.
- Built a **provider-agnostic AI service** with configurable adapters for **OpenAI, Gemini and Groq**, including provider fallback, rate-limit handling, retries, token/cost controls, API error handling and response validation. Reduced typical agent execution time from **30–60s to 6–18s** through caching and API/LLM optimization.
- Built an **HTML/AI personalization pipeline** for **10K–35K character webpages**, recursively splitting content into model-sized components while preserving structure for deterministic reconstruction and validation; reduced processing time from **30–50s to 6–12s** and extended the pipeline to image personalization.
- Investigated and resolved production incidents involving **AWS Lambda, SQS, Redis, PostgreSQL, API latency and EC2 memory/CPU saturation**; handled rollbacks and hotfixes using CloudWatch/application logs. Contributed to AWS cost optimization, Redis infrastructure optimization and resource-efficiency improvements.

Associate Software Developer, OpenSense Labs

July 2023 – April 2024

- Implemented consent-aware tracking and customer segmentation workflows using **Apache Unomi**, controlling data collection based on user consent and regional privacy requirements.
- Developed backend personalization and automation features and a **RAG proof of concept** using **Node.js, Python, OpenAI embeddings, Pinecone and GPT-3.5-turbo**, covering data processing, chunking, embeddings, vector search, KNN retrieval, cosine similarity and contextual retrieval.

Projects

Distributed Job Scheduler

Node.js, Redis, PostgreSQL

- Built a distributed job scheduler from scratch with **Redis queueing and PostgreSQL job state**, supporting scheduled execution, concurrent workers, retries, idempotency, failure recovery, visibility/state transitions and request rate limiting.

KragDB

C++

Building a database system from scratch to explore storage, data management and database internals.

Technical Skills

- Languages:** JavaScript, Python, SQL, C++
- Backend:** Node.js, Express.js, Fastify, REST APIs, Distributed Systems, Event-Driven Architecture, Queue Processing, Worker Systems, Async Pipelines
- Databases:** PostgreSQL, Redis, DynamoDB, MongoDB, Sequelize; SQL, Indexing, EXPLAIN ANALYZE, Transactions, Joins, Schema Design, Pagination
- Reliability:** Idempotency, Retries, Rate Limiting, Backpressure, Failure Recovery, Caching, TTL-Based Processing, Production Debugging
- AI & Integrations:** Claude Code, OpenAI, Gemini, Groq, OAuth 2.0, Google APIs, GA4, Webhooks, LLM Integration, Agentic AI, Tool Calling, RAG, Embeddings, Vector Search
- Cloud & Infrastructure:** AWS Lambda, SQS, API Gateway, S3, IAM, VPC, CloudWatch, EC2, Terraform, Jenkins
- Security:** JWT, OAuth 2.0, RBAC, API Authorization, IAM, Secrets Management, Encryption, Audit Logging

Education

Bachelor of Technology in Mechanical Engineering

July 2019 – June 2023

Ujjain Engineering College, Ujjain